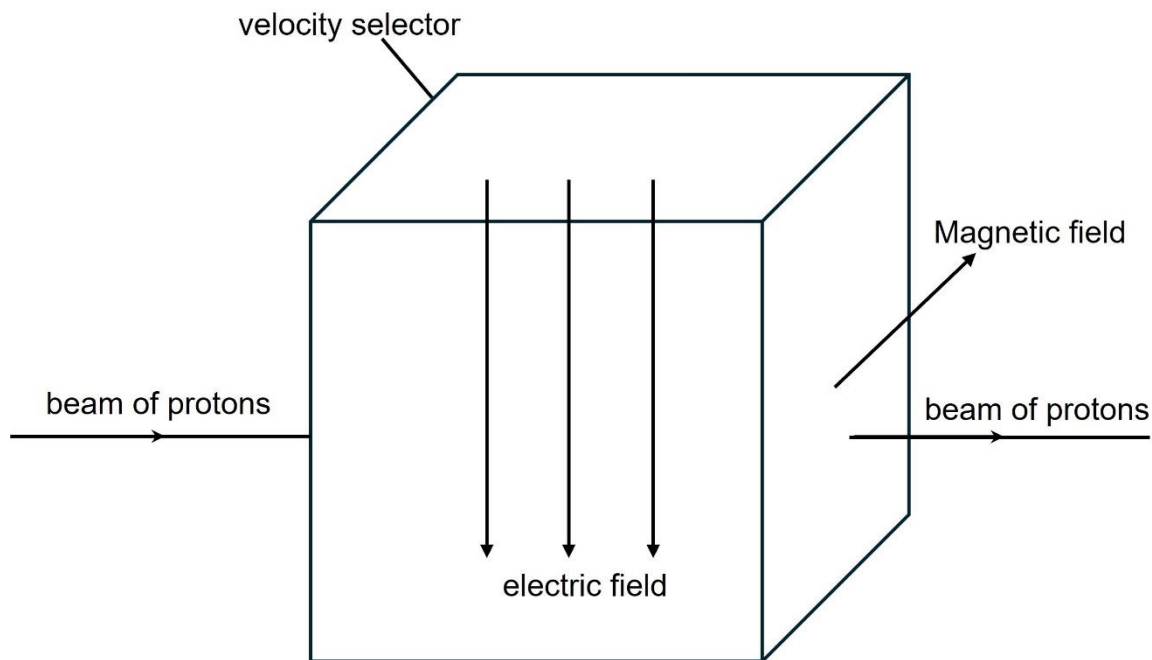


- 23** A beam of protons enters a velocity selector as shown. The electric field E acts vertically downwards, and the magnetic field B is directed into the page.



If only protons with a particular speed emerge undeflected, which change would result in the protons being deflected downwards?

- | | |
|----------|--|
| A | Increasing the electric field strength only |
| B | Increasing the magnetic field strength only |
| C | Increasing both E and B by the same factor |
| D | Increasing the proton's speed |

Answer: A

For no deflection, $E = vB$. Increasing E only makes the electric force greater compared to the magnetic force, causing downward deflection.

- 24** A rectangular coil of wire lies in a uniform magnetic field of 0.30 T . The field is perpendicular to the